Personality and achievement: A follow-up study

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Received: 10 March 2021
Revised: 23 September 2021
Accepted: 14 November 2021
Published: 12 April 2022

ABSTRACT

Background: The health and well-being of medical students along with their successful progression is an important concern for society. Studies across the world demonstrated a high prevalence of psychiatric morbidity and low self-esteem among medical students requires immediate attention and effective management. **Aim:** To compare the symptoms of depression, anxiety, and stress between the groups who have cleared all papers in the first attempt with those who have got supplementary at least in one paper during their MBBS course. **Materials and Methods:** Raven's Progressive Matrices Scale, Eysenck personality Questionnaire, Sinha's Anxiety Scale, Stress Reaction Check (SRC) List, and 16-Personality Factors questionnaires were used along with performance score was recorded for the objective of the study. **Results:** Nonparametric tests, Mann–Whitney and Wilcoxon are used and found that RSPM scores and EPQ Lie-scale scores are statistically different as the $P < 0.05$. Overall performance and psychological problems association are found out using Spearman's correlation test and it is found that Factor C and Tough poise under 16 PF are significantly correlated. **Conclusions:** The study concluded that there is a significant difference between the abstract reasoning between the two groups of students measured by RSPM. When the students are compared by taking gender as grouping variable the study concluded psychosis, lie scale value is significantly different in the two groups and the mean value is higher in the female group. It is also concluded that the stress level measured by SRC is significantly different and the mean value is higher in the female group. Furthermore, the global factors Tough poise and independence are found to be significantly different in two groups with 16 PF questionnaire.

Keywords: Academic performance, Mann–Whitney test, personality questionnaire, Spearman rank correlation

The health and well-being of medical students along with their successful progression through medical training is an important concern for society as their role is to care for the nation's sick and to promote public health. Previous studies suggest the educational process and course curriculum may have a negative impact on students' mental health which may lead to depression, anxiety, and stress among medical students as compared to students enrolled in other courses.[1] Long working hours and continuous work pressure sometimes may result in substance and alcohol abuse which may result in problems of depression, anxiety, and mood disorders.[2-5] Many medical students face different levels of consciousness, psychological stress and physiological stress, and points of stress can be interrelated with each other and be at different levels.[6,7] The prevalence of stress was higher during the first 3 years of medical study and especially among female students.[7] The Resident Service Committee of the Association of Program Directors in Internal Medicine divided the common stressors of residency into three categories; situational, personal, and professional.[8] Studies across the world demonstrated mental health problems in medical students viz psychiatric morbidity was found in 16% of medical students from the UK,[9] whereas 38.2% students

Access this article online

Quick Response Code:

Website: www.industrialpsychiatry.org

DOI: 10.4103/ipj.ipj_57_21

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How to cite this article: Srivastva R, Sabharwal A, Agrawal A, Anita, Vershney H, Srivastava Y. Personality and achievement: A follow-up study. Ind Psychiatry J 2022;31:19-25.

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were found to be affected with depressive symptoms from a study based on Brazil medical students.[10] In their 2013 study, Singh and Jha compared anxiety, optimism, and academic achievement among students of private medical and engineering colleges and found that 56% of medical students were high on anxiety scores as compared to 36% of engineering students. Further, their study concluded optimism was a significant predictor for both anxiety and academic achievement.[11] Rotenstein *et al.*, by using systematic review and meta-analysis, found that the estimated frequency of depression or its manifestations in medical students around the world was 27.2%.[12] Another study by Ahmed *et al.* in the United Arab of Emirates examined depression and anxiety in medical students, doctors, and members of the Medical College for Girls in Dubai. Their study concluded that 28.6% of medical undergraduates and 7.8% of medical staff had a high score for depression.[13]

Studies across the world demonstrated a high prevalence of psychiatric morbidity and low self-esteem among medical students which requires immediate attention and effective management.[14] Lievens *et al.* (2002) study suggested that personality traits are typical in medical students as compared to other students, and personality traits predict medical student performance in preclinical years. The study concluded that the higher scores on two dimensions extraversion and agreeableness defining the interpersonal dynamic of students are beneficial in future professional practice. Furthermore, conscientiousness(i.e.self-achievement and self-discipline) affects examination results as it predicted final scores in each preclinical year. Therefore, personality assessment may be a useful tool in student counseling and guidance.[10] Preventive mental health services, therefore, could be made an integral part of routine clinical services for medical students.

As most of the college students face borderline personality disorder, so diagnosis and intervention at the right time is a major challenge as time is highly variable. The criteria for diagnosing borderline personality disorder depends on the degree of emotional lability, interpersonal instability, and identity confusion are more typical in young adults. Various tests are used to measure general mental ability. The Raven's Progressive Matrices Test is used to measure basic cognitive functioning for different cultural, ethnic, and socioeconomic groups.[15] Eysenck personality questionnaire (EPQ) is used to study four dimensions of personality and has been used to assess the effect of cultural differences on personality. Using EPQ, Damodara's (2015) study established the relation of Neuroticism with anxiety, depression, and loss of behavioral and emotional control problems among the youth of Kerala, India.[16] To assess and compare the level of Anxiety of from the Aligarh Muslim University in Uttar Pradesh, Sinha's Anxiety Scale (SAS) was administered to find out the level of anxiety of the two genders' data. It was found that the two groups of students were not significantly different in respect of Anxiety at a 5% level of significance. Gavrilescu and Vizireanu (2017) study applied machine learning tools to analyze the facial features of individuals along with the Sixteen Personality factors questionnaire to evaluate different psychological characteristics.[17]

The main objectives of this paper are to compare the symptoms of depression, anxiety, and stress of (i) between the groups who have cleared all papers in the first attempt with those who have got supplementary at least in one paper during their MBBS course; and (ii) male and female groups. In this paper, we consider data sets representing the scores obtained by students with standardized psychological measures along with their performance records. Two nonparametric methods are used to compare the scores in different dimensions of the personality of two groups, in the above two cases. The relationship between performance and score obtained in different psychological tests for both the cases has been examined using Spearman rank correlation. The remainder of the paper is organized as follows. In section 2, material and methods have been discussed. Section 3 applies the tests to a primary data set of 80 medical students enrolled in Santosh Medical College, Ghaziabad, Uttar Pradesh, India in the year 2009. Some concluding remarks are made in section 4 under discussion.

MATERIALS AND METHODS

Materials

A follow-up study was conducted in 2009–2015 in Santosh Medical University, Ghaziabad, UP, India from the 1st year students enrolled in MBBS till their rotatory internship to assess their mental health. This study included 80 willing medical students, out of 100 enrolled for the course. Before starting the actual assessment, all students were interviewed to know their previous psychological profile, their academic background, any history of mental illness along with their mental status examinations. Data were collected using standardized psychological measures, namely; (i) Raven's Progressive Matrices [8] scale consisting of 60 problems divided into five sets of 12 in each set; (ii) EPQ 90-item EPQ questionnaire having 24 "Psychoticism" items, 21 "Extraversion" items, 24 "Neuroticism" items and 21 "Lie-scale items"[18] (iii) Sinha's Anxiety Scale (SAS) which is a self-administered inventory having a total of 100 questions related to Psychological manifestations, Ambition, Future, Family, Relations, Friendship, Love, Health, virtue and war, Sinha, D;[19] (iv) Stress Reaction Check List (SRC; SRC-It is questionnaire consisting of 20 questions designed by

Srivastva, *et al.* in 2011, Department of Clinical Psychology, Santosh Medical University, Ghaziabad, UP, India. Unpublished), a self-administered inventory having a total of 20 questions and; (v) 16-Personality Factors (16-PF)), a self-report Questionnaire containing 185 multiple-choice items. The sixteen primary scales and five global scales are Extraversion, Anxiety Neuroticism, Tough-Mindedness, Independence, and Self-Contro.[20,21] Along with the above information, we had used self-designed Performa for recording their academic performances during each year of course.

Methods

We compared the scores obtained in different dimensions of personality using Mann–Whitney and Wilcoxon tests to compare groups; (i) students who completed their medical school successfully with those students who got supplementary in at least 1 year during their course and; (ii) on the basis of grouping variable gender. Further, we applied Spearman rank correlation to find the association of performance with personality score obtained by using different tools (discussed under the Material section).

RESULTS

This follow-up study was conducted on a group of 100 medical students who belonged to the same batch of 2009 in Santosh Medical University. There were 46 boys and 54 girls. After completion of their 1st year, they were asked for their willingness to participate in the study. A total of 80 students, 35 boys and 45 girls consented to participate in the study. Among these 80 students, 48 (18 boys and 30 girls) cleared all their examinations while the rest 32 (17 boys and 15 girls) had supplementary in one or more papers. These 80 students were monitored for their academic performance through the rest of their course. Furthemore, they were subject to standard investigation tools, namely (1) Ravens Standard Progressive Matrices (RSPM);[8] (2) EPQ;[18] (3) Sinha Anxiety Scale (SAS);[19] (4) SRC List and (5) 16-PF.[20,21]

Table 1 presents the descriptive statistics of boys and girls in respect of all the dimensions of all the five questionnaires categorized according to their examination clearance status.

RSPM score for students who have cleared all exams is much higher (51.26) as compared to those who have failed in one or more papers (47.19). High variability is there in SAS scores, in students who got supplementary, EPQ mean and median values are higher in students who got supplementary except for neuroticism. Similar results are being shown by SAS and SRC scores where higher means are being accompanied by higher medians. 16PF mean scores of Tough poise and independent dimensions are higher for those who cleared their examination successfully at each year.

After obtaining the descriptive statistics we compared these two groups on different dimensions of personality, anxiety, and stress by using two nonparametric tests. Nonparametric tests were used since for most of cases assumption of normality was rejected. The following hypotheses were set:

H_0: There is no significant difference between personality dimension, intelligence, and stress levels of students classified according to the examination clearance status.

H_1: There is a significant difference between personality dimension, intelligence, and stress levels of students classified according to the exam clearance status.

The hypotheses were tested using Mann–Whitney U test and Wilcoxon tests. The results are presented in Table 2.

At a 5% level of significance, only RSPM and EPQ Lie-scale were statistically different. However, at a 10% level of significance EPQ Extraversion, 16PF_A and 16PF_Q1 were also differ significantly.

The procedure was repeated for gender. Tables 3 and 4 present the descriptive statistics and non-parametric test result for males and females.

RSPM scores were similar for the two groups. As for EPQ scores, males had higher average scores on Psychoticism and Extraversion but for Neuroticism and Lie-scales, females had a higher average score. Sinha anxiety scale mean score of females (30.93) is much higher than an average of 24.29 for males and comparison holds good for anxiety score measured with 16 PF. Surprising, the Tough poise average of the female group (11.28) is much higher than the male group average score (2.23).

The two groups were compared statistically using the following hypothesis

H_0: There is no significant difference between personality dimension, intelligence, and stress levels of students taking gender as grouping variable.

H_1: There is a significant difference between personality dimension, intelligence, and stress levels of students taking gender as grouping variable.

It was found that the two groups were significantly different in respect of Psychoticism, Lie scale of EPQ,

Table 1: Descriptive statistics for personality dimension, intelligence and stress levels of students classified according to the exam clearance status

	Examination status - Complete (size=47)					Examination status - Supplementary (size=33)				
Dimension	Mean	SD	Median	Minimum	Maximum	Mean	SD	Median	Minimum	Maximum
RSPM	51.26	6.1	53	28	60	47.19	5.36	49	31	53
Personality_P	5.83	3.52	6	0	14	6.19	3.6	7	1	15
Personality_E	12.72	4.27	13	2	21	14.34	2.9	14	9	19
Personality_N	12.55	5.09	13	2	22	11.06	4.7	11	1	19
Personality_L	10.19	4.48	11	0	19	12.41	4.19	12	3	20
SAS	28.87	15.77	29	1	82	26.69	17.89	25.5	1	68
SRC	17.77	7.89	16	3	43	14.75	5.67	15	0	25
16Pf_A	9.62	2.95	10	2	16	10.44	2.86	11	2	16
16Pf_C	13.91	4.39	13	2	24	13.03	3.37	13.5	4	19
16Pf_E	11.81	3.52	12	5	20	12.53	4.04	13	5	20
16Pf_F	11.34	3.87	12	4	21	12.56	4	12	4	20
16Pf_G	12.21	3.64	13	4	19	11.44	3.33	12	3	17
16Pf_H	12.47	4.83	12	2	22	12.94	3.71	13	3	19
16Pf_I	11.04	4	11	2	22	10.97	3.85	11	4	20
16Pf_L	10.19	3.38	10	2	17	10.62	3.22	10	3	20
16Pf_M	10.28	3.42	10	4	18	11.38	3.54	11	5	22
16Pf_N	11.45	3.44	12	1	18	10.88	3.72	11	4	18
16Pf_O	12.32	4.7	12	1	25	13.12	3.52	12	8	24
16Pf_Q1	8.98	3.22	8	3	15	10.19	3.15	10	2	17
16Pf_Q2	10.3	3.83	10	2	22	9.59	3.1	10	4	16
16Pf_Q3	11.55	3.26	12	5	18	11.09	2.94	11	6	18
16Pf_Q4	12.4	5.67	13	0	21	12.38	4.38	12	2	22
16Pf_Exver	10.55	3.34	11	3.1	16.1	11.42	2.72	11.35	5	18.2
16Pf_Anxiety	8.27	4.45	8.9	−2.3	20.9	8	3	7.55	2.6	17.4
16Pf_Tpoise	7.95	5.21	7.6	−3.1	17.5	6.85	6.99	7.6	−16.1	17.7
16Pf_Indepen	9.86	3.64	9.9	4.2	21.8	8.98	4.07	9.55	1.2	19.4

SD – Standard deviation, SAS – Sinha's Anxiety Scale, SRC – Stress reaction check list, RSPM – Ravens standard progressive matrices, 16-PF – 16-personality factors

Table 2: Nonparametric tests to check the significance of differences in scores of Ravens standard progressive matrices, Eysenck personality questionnaire, Sinha's Anxiety Scale, stress reaction check list and 16-personality factors for two groups of students categorized according to their exam performance

Test statistic	Group	RSPM	P_P	P_E	P_N	P_L	SAS	SRC	PF16_A
Mann-Whitney U	Examination status	336	705	579	623.5	547	688	611.5	582
Wilcoxon W		864	1833	1707	1151.5	1675	1216	1139.5	1710
Z		−4.17	−0.472	−1.735	−1.287	−2.052	−0.639	−1.407	−1.711
Asymptotic significance (two-tailed)		<0.001	0.637	0.083	0.198	0.04	0.523	0.159	0.087
Test statistic	**PF16_C**	**PF16_E**	**PF16_F**	**PF16_G**	**PF16_H**	**PF16_I**	**PF16_L**	**PF16_M**	**PF16_N**
Mann-Whitney U	691.5	668	641	637.5	687	749	696.5	654.5	647
Wilcoxon W	1219.5	1796	1769	1165.5	1815	1877	1824.5	1782.5	1175
Z	−0.607	−0.842	−1.116	−1.15	−0.652	−0.03	−0.558	−0.985	−1.059
Asymptotic significance (two-tailed)	0.544	0.4	0.265	0.25	0.514	0.976	0.577	0.325	0.29
Test statistic	**PF16_O**	**PF16_Q1**	**PF16_Q2**	**PF16_Q3**	**PF16_Q4**	**PF16_EXT**	**PF16_ANX**	**PF16_TOUGH**	**PF16_IND**
Mann-Whitney U	688	578.5	681	677.5	732	628.5	685	689	716
Wilcoxon W	1816	1706.5	1209	1205.5	1260	1756.5	1213	1217	1244
Z	−0.642	−1.743	−0.713	−0.748	−0.2	−1.234	−0.669	−0.629	−0.36
Asymptotic significance (two-tailed)	0.521	0.081	0.476	0.454	0.841	0.217	0.503	0.529	0.719

SAS – Sinha's Anxiety Scale; SRC – Stress reaction check list; RSPM – Ravens standard progressive matrices; PF-16 – 16-personality factors

SRC, 16PF_O, 16PF_Tough and 16PF_IND at a 5% level of significance. At 10% level of significance, we observed that 16PF_Q2, 16 PF_M, and 16PF_ANX are significantly different.

Table 3: Descriptive statistics for personality dimension, intelligence and stress levels of students classified according to gender

	Male group (size=35)					Female group (size=45)				
Dimension	Mean	SD	Median	Minimum	Maximum	Mean	SD	Median	Minimum	Maximum
RSPM	49.91	6.26	51	31	60	49.36	6.06	50.5	28	59
Personality_P	7.8	3.4	8	1	15	4.52	2.94	3	0	11
Personality_E	14.26	3.35	14	5	21	12.68	4.1	13	2	20
Personality_N	10.94	5.47	12	1	22	12.75	4.43	13	5	21
Personality_L	9.94	4.9	10	0	20	12	3.92	12	3	19
SAS	24.29	15.67	25	1	54	30.93	16.88	29.5	1	82
SRC	13.4	6.07	14	0	32	19.05	7.09	17	10	43
16Pf_A	9.86	3.18	10	2	16	10.02	2.74	10	5	16
16Pf_C	14.2	3.89	14	7	24	13.05	4.07	13	2	22
16Pf_E	12.66	3.85	13	7	20	11.66	3.61	12	5	20
16Pf_F	12.26	3.91	12	6	20	11.5	3.99	12	4	21
16Pf_G	11.69	3.76	12	3	19	12.07	3.34	13	4	17
16Pf_H	13.2	4.12	14	3	22	12.23	4.59	12	2	21
16Pf_I	10.49	3.94	10	4	22	11.43	3.89	12	2	18
16Pf_L	10.66	3.24	10	2	20	10.14	3.37	10	3	17
16Pf_M	11.66	4.01	12	4	22	9.98	2.85	10	4	18
16Pf_N	10.11	3.77	11	1	17	12.09	3.12	12	5	18
16Pf_O	11.37	3.87	11	1	18	13.66	4.31	14	5	25
16Pf_Q1	10.14	3.58	10	2	17	8.93	2.84	9	3	15
16Pf_Q2	10.57	2.99	11	4	16	9.57	3.91	9	2	22
16Pf_Q3	10.77	2.91	11	6	16	11.84	3.24	12	5	18
16Pf_Q4	11.83	4.83	12	0	20	12.84	5.41	14	2	22
16Pf_Exver	11.27	2.92	11.3	5	16.1	10.61	3.27	10.75	3.1	18.2
16Pf_Anxiety	7.19	3.42	7.4	−2	15.7	8.86	4.19	9.05	0.4	20.9
16Pf_Tpoise	2.23	4.33	3.1	−16	7.6	11.28	3.53	11.35	3.7	17.7
16Pf_Indepen	10.45	4.12	10.7	1.2	19.4	8.88	3.46	8.6	1.4	21.8

SD – Standard deviation; SAS – Sinha's Anxiety Scale; SRC – Stress reaction check list; RSPM – Ravens standard progressive matrices; 16-PF – 16-personality factors

Table 4: Nonparametric tests to check the significance of differences in scores of Ravens standard progressive matrices, Eysenck personality questionnaire, Sinha's Anxiety Scale, stress reaction check list and 16 personality factors for two groups of students categorized according to gender

Test statistic	Group	RSPM	P_P	P_E	P_N	P_L	SAS	SRC	PF16_A
Mann-Whitney U	Gender	706.5	361	607	624	572	605.5	406.5	741
Wilcoxon W		1696.5	1351	1597	1254	1202	1235.5	1036.5	1731
Z		−0.629	−4.06	−1.616	−1.445	−1.959	−1.624	−3.598	−0.288
Asymptotic significance (two-tailed)		0.529	<0.001	0.106	0.148	0.05	0.104	<0.001	0.773
Test statistic	**PF16_C**	**PF16_E**	**PF16_F**	**PF16_G**	**PF16_H**	**PF16_I**	**PF16_L**	**PF16_M**	**PF16_N**
Mann-Whitney U	671.5	669	727.5	706.5	676.5	618.5	701	582	532
Wilcoxon W	1661.5	1659	1717.5	1336.5	1667	1248.5	1691	1572	1162
Z	−0.976	−1.001	−0.422	−0.63	−0.927	−1.502	−0.686	−1.877	−2.373
Asymptotic significance (two-tailed)	0.329	0.317	0.673	0.529	0.354	0.133	0.493	0.061	0.018
Test statistic	**PF16_O**	**PF16_Q1**	**PF16_Q2**	**PF16_Q3**	**PF16_Q4**	**PF16_EXT**	**PF16_ANX**	**PF16_TOUGH**	**PF16_IND**
Mann-Whitney U	540	609	582.5	626	683	678.5	584	31	531.5
Wilcoxon W	1170	1599	1572.5	1256	1313	1668.5	1214	661	1521.5
Z	−2.279	−1.598	−1.86	−1.429	−0.861	−0.903	−1.836	−7.294	−2.354
Asymptotic significance (two-tailed)	0.023	0.11	0.063	0.153	0.389	0.366	0.066	<0.001	0.019

SAS – Sinha's Anxiety Scale; SRC – Stress reaction check list; RSPM – Ravens standard progressive matrices; 16-PF – 16-personality factors

As the objective of the study is to observe the relationship between different personality aspects and the performance of students in their examinations, the association is computed using the following hypothesis

Table 5: Correlations of Ravens standard progressive matrices, Eysenck personality questionnaire, Sinha's Anxiety Scale, stress reaction check list and 16 personality factors scores with the average marks of students in 4 years

	Average marks in 4 years (performance)								
Factor	RSPM	P_P	P_E	P_N	P_L	SAS	SRC	16Pf_A	16Pf_C
R	0.1515	−0.0118	0.0536	0.1759	−0.1223	0.13	0.1565	0.0416	0.2512
P	0.139	0.396	0.353	0.093	0.236	0.186	0.128	0.370	0.016
Factor	16Pf_E	16Pf_F	16Pf_G	16Pf_H	16Pf_I	16Pf_L	16Pf_M	16Pf_N	16Pf_O
R	−0.1685	0.0409	0.1785	0.1538	−0.0965	−0.1275	−0.1534	0.039	0.0272
P	0.154	0.371	0.088	0.134	0.285	0.226	0.179	0.374	0.386
Factor	16Pf_Q1	16Pf_Q2	16Pf_Q3	16Pf_Q4	16Pf_Exver	16Pf_Anx	16Pf_Tpoise	16Pf_Ind	
R	0.0424	−0.0844	0.0806	0.1217	0.1152	−0.0048	0.2827	−0.0959	
P	0.369	0.307	0.301	0.205	0.221	0.397	0.006	0.286	

SAS – Sinha's Anxiety Scale; SRC – Stress reaction check list; RSPM – Ravens standard progressive matrices; 16-PF – 16-personality factors

H_0: There is no correlation between personality dimension, intelligence, and stress levels of students with their overall performance in the examination.

H_1: There is a significant correlation between personality dimension, intelligence, and stress levels of students with their overall performance in the examination.

The Spearman's rank correlations were calculated between various scores and average marks obtained by students in 4 years along with *P* value. The results are presented in Table 5.

Psychoticism and Lie scales scores had negative correlations with the performance and so were 16PF E, I, L, M, Q2, Q3 and Anxiety. At 5% level of significance C and global dimension, Tough poise under 16 PF are significantly correlated with the performance of the student.

CONCLUSIONS

Comparing the groups of students by examination status: Group (i) of students who successfully completed the examination is compared with the group (ii) in which respondents got supplementary at least in 1 year. As RSPM scores measure abstract reasoning and it is evident from Tables 1 and 2, the two groups are significantly different and mean score of the group (i) is higher. EPQ is a reliable tool which measures the four dimensions of personality, of which the Lie scale value is significantly different in two groups and the mean value is higher in group (ii). These findings are consistent with the previous studies. Furthermore, extraversion dimension is significant at 10% level of significance. There is no significant difference among other factors; anxiety, stress, and other personality factors measured through SAS, SRC, and 16 PF. Further, results shown in Table 5 suggest that there is no significant linear association found between overall performance and psychological problems except factor C and global dimension Tough poise under 16 PF. Factor C and global dimension Tough poise under 16 PF are found to be significantly correlated.

Comparing the groups gender wise: As is evident from Tables 3 and 4, RSPM scores are not significantly different for the two groups, and hence the level of abstract reasoning is almost the same for the two genders. EPQ tool which measures the four dimensions of personality, out of four dimensions, psychosis, lie scale value is significantly different in two groups and the mean value is higher in the female group. It is also concluded that the stress level measured by SRC is significantly different and mean value is higher in the female group Furthermore, the global factors Tough poise and independence are found to be significantly different in the two groups with 16 PF questionnaire.

As this is a follow-up study which is based on one batch of students and there is no other study group to prove the objective of the study can be taken as limitation of the study.

Financial support and sponsorship
Nil.

Conflicts of interest
There are no conflicts of interest.

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